👩‍🏫 Staff Guide · Biology A · Unit 3 Biofuel & Energy (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 2 Algae Biofuel: A3 Matter & Energy (Characteristics of Matter and Energy, Energy Transfer & Transformation, Crash Course Chemistry, Bill Nye Energy), A5 Making Lipids, A6 Making Biofuel + Combustion Lab, A7 Energy Reactions, A8 Carbon Cycle, A10 Global Connections, A2.10 Algae Biofuel Review, Algae Biofuel Final Exam")	**Level 2 piece**
A3 Matter & Energy worksheets + Crash Course Chemistry + Bill Nye Energy	Notes part A + slides 1-6 (her two A3 worksheets grouped as one lesson)
A5 Making Lipids (Lego model)	Notes part B + lipid slide
A6 Making Biofuel + Combustion Lab (10 pts)	Biodiesel observer packet (🔬 Labs & Projects)
A7 Energy Reactions	Energy-changes packet + notes part D
A8 Carbon Cycle (+ webquest)	Notes part E + carbon-cycle order game
A10 Global Connections	🚧 Under construction (her topics are only pictures)
Algae Test review + Algae Biofuel Final Exam / paper Algae Test	Modified unit test + key
Class slides	Deck: 24 slides, Words to Know first, hold-up-1-2-3 checks, recap
Her worksheets (grouped into lessons)	Guided notes: 21 items + key
Class quiz / notes check	Quiz: 13 questions + key
Vocabulary	13 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no "all of the above"
- Picture anchors: ⚡ energy = makes things move, 🪨 stored vs 🏃 moving, 🧈 lipid = oil
- Level 1 uses no chemical formulas; Level 2 uses word equations only

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part

4. What to grade

- Unit test: same weight as the class Algae test, scaled
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **A10 Global Connections**: Her 24 connection topics are only in a picture PDF (no text). We need the topic list to build a picture reading.
- **Carbon cycle webquest**: Title only in Canvas; no file or link.
- **Biofuel sorter**: Title only in Canvas; no file.
- **Paper Algae Test (32 pts)**: Not in the export. Our test is built from her Final Exam quiz and review.

6. Photo credits (open licenses)

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👩‍🏫 Staff Guide · Making & Burning Biodiesel (Observer Jobs) (Level 2)

Biology A · Unit 3 · goes with the class “A6 Making Biofuel, Day 2 Burning Biofuel, Combustion Lab report” (5 + 5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Observer card**: A6 Making Biofuel + Day 2 burning + Combustion Lab: our students observe and record; staff or the teacher handle methanol, the catalyst and the flame. 8 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “What color is the oil?”
- “Is the water hotter or colder now?”
- “Point to the top layer.”

3. Modifications

- Observer job only (no chemicals, no flame)
- Data table given; no heat-of-combustion math (kJ/g)
- Energy question with a word bank

4. What to grade (5 + 5 pts)

- Same 5 + 5 points: data table, energy question, reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Ms. Dunham said in Canvas the teacher adds the catalyst. Please confirm our students' observer roles and where the safety line goes.

6. Photo credits (open licenses)

Biodiesel.JPG — Shizhao拍摄, CC BY-SA 3.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Energy Changes: Hot or Cold? (Level 2)

Biology A · Unit 3 · goes with the class “A7 Energy Reactions” (5 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A7 Energy Reactions: Investigation 1 (baking soda + vinegar) and Investigation 2 (baking soda solution + calcium chloride). 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Is the number going up or down?”
- “Hot or cold?”
- “Which one is like a hand warmer?”

3. Modifications

- Table given; energy-arrow drawing replaced by up/down + endo/exo choice
- Word bank for the analysis

4. What to grade (5 pts)

- Same 5 points: data table + endo/exo answers + reflection
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

⛽ Biofuel & Energy notes (A3, A5, A6, A7, A8)

Biology A · Unit 3 Biofuel & Energy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Matter and energy (A3)

1. Matter has _____ and takes up space.
2. Energy has no mass. It makes things move or _____.
3. Match the energy to the example:

Potential	food and fuel
Kinetic	heat
Chemical	water behind a dam
Thermal	a rolling boulder

4. Energy moving to a new place is a _____. Energy changing form is a transformation.

Word bank: **change** **mass** **transfer**

B. Lipids (A5)

5. Fats and oils are called _____.
6. A fat is made of glycerol and three fatty _____.
7. Which stores more energy per gram? (circle 1)

lipids (fats) sugar

Word bank: **acids** **lipids**

C. Making biofuel (A6)

8. Biodiesel is made from plant or algae _____.
9. Biodiesel is _____ because plants grow back fast.
10. Number the energy changes when a bus burns biodiesel (1-3):

_____ heat when it burns

_____ chemical energy in the fuel

_____ mechanical energy moves the bus

Word bank: **oil** **renewable**

D. Energy reactions (A7)

11. A reaction that gets hot is _____.

12. A reaction that gets cold is _____.

13. Baking soda + vinegar made the cup… (circle 1)

colder hotter

Word bank: **endothermic** **exothermic**

E. Carbon cycle (A8)

14. Coal, oil, and natural gas are _____ fuels.

15. Plants and algae take carbon dioxide out of the _____.

16. Which do NOT put carbon dioxide into the air? (circle 1)

photosynthesis burning breathing

Word bank: **air** **fossil**

Level 2 version of the gen ed "A3 Characteristics of Matter and Energy, A5 Making Lipids, A6 Making Biofuel, A7 Energy Reactions, A8 Carbon Cycle."

📖 Biofuel & Energy: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Energy

The ability to make things move or change. Energy has no mass.



Matter

Anything that has mass and takes up space (oil, water, air).



Potential energy

Stored energy, like a rock at the top of a hill or fuel in a tank.



Kinetic energy

The energy of motion, like a rock rolling down a hill.

Chemical energy

Energy stored in the bonds of food and fuel. Living things use it.

Thermal energy

Heat energy. Burning fuel gives off thermal energy.



Lipid

Fats and oils. They store about 2.5 times more energy than sugar, gram for gram.



Biodiesel

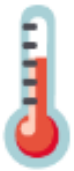
A renewable fuel made from plant or algae oil. It runs in diesel engines.

Renewable

A fuel source we can make again in a short time (plants, algae).

Fossil fuel

Coal, oil, and natural gas. Made over millions of years. Non-renewable.



Exothermic

A reaction that releases heat. The temperature goes up.



Endothermic

A reaction that takes in heat. The temperature goes down.



Carbon cycle

Carbon moves from air to plants to animals and fuel, then back to the air.

🛢️ Making & Burning Biodiesel (Observer Jobs)

Biology A · Unit 3 · Level 2



Name: _____

A6 Making Biofuel + Day 2 burning + Combustion Lab: our students observe and record; staff or the teacher handle methanol, the catalyst and the flame.

🧺 What I need (observer)

- ☐ 🥽 **1** pair of goggles (on the whole time)
- ☐ 📋 **1** observer card + pencil
- ☐ ⏱️ **1** timer or clock
- ☐ 📷 **1** tablet for photos (if allowed)

🌱 Steps: check each box when you finish

1. 🌻 **Observe** the canola oil: color, thickness, smell (wave the air to your nose). ☐
2. 🧪 **Watch** the teacher add methanol and the catalyst (teacher only). ☐
3. ⏱️ **Time** 10 minutes of shaking. Record. ☐
4. 🥞 **Day 2**: record the 2 layers. Biodiesel is on top, glycerol on the bottom. ☐
5. 🌡️ **Record** the start temperature of the water in the can. ☐
6. 🔥 **Watch** the biodiesel burn for 5 minutes (teacher lights it). ☐
7. 🌡️ **Record** the highest water temperature. Find the change. ☐
8. 📝 **Answer**: what forms of energy did you see? (chemical → heat and light) ☐

📋 Data table

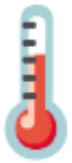
What	Before	After
Oil / biodiesel color		
Thickness (thick / thin)		
Water temperature (°C)		
Temperature change (°C)		

💡 *This lab is for watching and recording. Our students never handle methanol, potassium hydroxide, or the flame.*

🚧 **Under construction:** Ms. Dunham said in Canvas the teacher adds the catalyst. Please confirm our students' observer roles and where the safety line goes. *Check with Ms. Dunham.*

🌡️ Energy Changes: Hot or Cold?

Biology A · Unit 3 · Level 2



Name: _____

A7 Energy Reactions: Investigation 1 (baking soda + vinegar) and Investigation 2 (baking soda solution + calcium chloride).

🧺 Investigation 1

- ☐ 🍶 **1 cup** vinegar
- ☐ 🧂 **1 cup** baking soda
- ☐ 🌡️ **1** thermometer

🧺 Investigation 2

- ☐ 💧 **1 cup** baking soda solution
- ☐ 🧂 **1 cup** calcium chloride (staff pours)
- ☐ 🌡️ **1** thermometer

🌱 Steps: check each box when you finish

1. 🌡️ **Investigation 1**: put the thermometer in the vinegar. Record the start temp. ☐
2. 🧂 **Add** all the baking soda. Watch it fizz. ☐
3. 📉 **Record** the end temp when it stops changing. Did it go up or down? ☐
4. 🌡️ **Investigation 2**: thermometer in the baking soda solution. Record the start temp. ☐
5. 🧂 Staff **adds** the calcium chloride. ☐
6. 📈 **Record** the highest temp. ☐
7. 🧠 **Decide**: temperature down = endothermic; temperature up = exothermic. ☐

📋 Data table

Investigation	Start °C	End °C	Up or down?	Endo or exo?
1 Vinegar + baking soda				
2 Baking soda solution + calcium chloride				

🛢️ Making & Burning Biodiesel (Observer Jobs) · Prep Sheet

Biology A · Unit 3 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
⏱️ **Timer**	Starts the timer: 10 minutes of shaking (Day 1), 5 minutes of burning (Day 2). Calls out the time.	
📝 **Recorder**	Writes the oil color, thickness, and the water temperatures in the table.	
📷 **Photo taker**	Takes a photo before and after: oil, two layers, the flame (from a safe spot).	
👀 **Color checker**	Says what color the oil and biodiesel are, and if there are 2 layers.	

🧰 Tools and supplies: check when you have it

- ☐ 🥽 **Goggles** everyone, the whole lab
- ☐ ⏱️ **Timer** shake 10 minutes / burn 5 minutes
- ☐ 🌡️ **Thermometer** read the water temperature
- ☐ 📋 **Data table** on the observer card

⚠️ Safety and manners focus

🥽 Goggles on the whole lab. Stay behind the safety line.

🧪 Methanol and potassium hydroxide are poison and burn skin. Only the teacher touches them.

🔥 Only the teacher lights and puts out the flame. Long hair tied back.

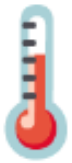
🧼 Wash hands after the lab. No food or drinks at the lab table.

✅ Before you start

☐ Goggles on ☐ I know my job ☐ Safety line on the floor ☐ Hands free (no food)

🌡️ Energy Changes: Hot or Cold? · Prep Sheet

Biology A · Unit 3 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🌡️ **Reader**	Reads the thermometer at the start and the end.	
🥄 **Adder**	Adds the baking soda to the vinegar (Investigation 1).	
📝 **Recorder**	Writes the start and end temperatures.	
🧽 **Cleaner**	Wipes the table and dumps cups where the teacher says.	

🧰 Tools and supplies: check when you have it

- ☐ 🌡️ **Thermometer** read before and after
- ☐ 🥽 **Goggles** everyone
- ☐ 🧤 **Gloves** for the calcium chloride cup
- ☐ 🧻 **Paper towels** for spills

⚠️ Safety and manners focus

- 🥽 Goggles on. Vinegar and calcium chloride can sting eyes.
- 🧤 Calcium chloride gets hot. Staff pours it; do not touch it.
- 🚫 Do not taste anything in the lab.
- 🧼 Wash hands after.

✅ Before you start

☐ Goggles on ☐ Thermometer in the cup ☐ Start number written

🛢️ Making & Burning Biodiesel (Observer Jobs) · Reflection

Biology A · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Oil before ☐ Two layers ☐ Burning (from the line) ☐ Thermometer reading

✅ Skills I used today (check all you did)

☐ 👀 I stayed behind the safety line

☐ 📝 I recorded color, layers, and temperatures

☐ 🤝 I did my observer job

☐ 🥽 I wore goggles the whole lab

⭐ What forms of energy did you see when the biodiesel burned?

1. _____
2. _____

Ideas: **chemical** **heat** **light**

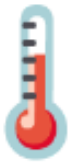
♻️ Why is biodiesel called renewable?

1. _____
2. _____

Ideas: **plants grow back** **made from oil** **short time**

🌡️ Energy Changes: Hot or Cold? · Reflection

Biology A · Unit 3 · Level 2



Name: _____

📸 My photos / proof

☐ Cup 1 data ☐ Cup 2 data ☐ Clean station

✅ Skills I used today (check all you did)

☐ 🌡️ I read and recorded the temperatures

☐ 🧠 I named endothermic or exothermic

☐ 🤝 I did my role

☐ 🥽 I wore goggles and did not taste

⭐ Name one thing in real life that gets hot or cold from a reaction.

1. _____
2. _____

Ideas: **hand warmer** **cold pack** **burning fuel**

🔧 How do you know a reaction is exothermic?

1. _____
2. _____

Ideas: **temperature goes up** **heat released**

🚧 Biofuel Sorter

Biology A · Unit 3 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Canvas has only the title. We do not know what students sort. *Check with Ms. Dunham.*

What we have so far

The matching game sorts renewable and non-renewable fuels for now.

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

✅ Biofuel & Energy Quiz

Biology A · Unit 3 · Level 2 (modified). 13 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⚡ Energy has mass.

A. True | B. False | C. Only light has mass

2. 🧱 Matter takes up space.

A. False | B. Only solids do | C. True

3. 🍎 What type of energy can all living things use directly?

A. Chemical energy | B. Light energy | C. Mechanical energy

4. 🪨 A boulder at the edge of a cliff has…

A. Kinetic energy | B. No energy | C. Potential energy

5. 🧈 Gram for gram, lipids give more energy than glucose because…

A. They have more high-energy C-H and C-C bonds | B. They are lighter | C. They have more oxygen

6. 🐣 A good fuel organism makes ___ offspring each year.

A. Few (low) | B. Many (high) | C. Zero

7. 🌾 A good fuel organism needs ___ fertilizer.

A. A lot (high) | B. Little (low) | C. Only salt

8. 🚌 Burning biodiesel: chemical energy → ___ → mechanical.

A. Heat | B. Light | C. Matter

9. 💨 When fuel burns, it uses oxygen and releases…

A. Oxygen and sugar | B. Only smoke | C. Water and carbon dioxide

10. 🔥 In an exothermic reaction, energy is…

A. Released (it gets hot) | B. Taken in (it gets cold) | C. Not changed

11. ♻️ Which fuel is a renewable biofuel?

A. Coal | B. Gasoline | C. Ethanol

12. 🌿 Which process does NOT put carbon dioxide into the air?

A. Burning | B. Photosynthesis | C. Breathing

13. 🔄 Algae biofuel keeps the carbon cycle balanced because carbon dioxide is…

A. Only made | B. Used and made | C. Only used

🚧 Carbon Cycle Webquest

Biology A · Unit 3 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Canvas has only the title. There is no file or website link. *Check with Ms. Dunham.*

What we have so far

Use the carbon-cycle order game and notes part E for now.

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

🚧 A10 Global Connections

Biology A · Unit 3 · Level 2 · Under construction

Name: _____

🚧 **Under construction:** Her 24 Global Connections topics are only pictures in the export (no words). We need the list of topics to make the picture reading. *Check with Ms. Dunham.*

What we have so far

The worksheet frame: for each connection write the title, what it is, and how I can help.

Staff: until this is ready

Have the student do the matching Level 2 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

⛽ Biofuel & Energy notes (A3, A5, A6, A7, A8) · ANSWER KEY

Biology A · Unit 3 Biofuel & Energy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Matter and energy (A3)

1. Matter has **mass** and takes up space.
2. Energy has no mass. It makes things move or **change**.
3. Match the energy to the example:

(Key: Potential → water behind a dam · Kinetic → a rolling boulder · Chemical → food and fuel · Thermal → heat)

Potential	food and fuel
Kinetic	heat
Chemical	water behind a dam
Thermal	a rolling boulder

4. Energy moving to a new place is a **transfer**. Energy changing form is a transformation.

Word bank: change · mass · transfer

B. Lipids (A5)

5. Fats and oils are called **lipids**.
6. A fat is made of glycerol and three fatty **acids**.
7. Which stores more energy per gram? (circle 1)

lipids (fats) · sugar

Word bank: acids · lipids

C. Making biofuel (A6)

8. Biodiesel is made from plant or algae **oil**.
9. Biodiesel is **renewable** because plants grow back fast.
10. Number the energy changes when a bus burns biodiesel (1-3):

2 heat when it burns

1 chemical energy in the fuel

3 mechanical energy moves the bus

Word bank: oil · renewable

D. Energy reactions (A7)

11. A reaction that gets hot is **exothermic** .

12. A reaction that gets cold is **endothermic** .

13. Baking soda + vinegar made the cup… (circle 1)

colder hotter

Word bank: **endothermic** **exothermic**

E. Carbon cycle (A8)

14. Coal, oil, and natural gas are **fossil** fuels.

15. Plants and algae take carbon dioxide out of the **air** .

16. Which do NOT put carbon dioxide into the air? (circle 1)

photosynthesis burning breathing

Word bank: **air** **fossil**

Level 2 version of the gen ed "A3 Characteristics of Matter and Energy, A5 Making Lipids, A6 Making Biofuel, A7 Energy Reactions, A8 Carbon Cycle."

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

✅ Biofuel & Energy Quiz · ANSWER KEY

Biology A · Unit 3 · Level 2 (modified). 13 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. ⚡ Energy has mass. Gen ed #9

A. True | **B. False** | C. Only light has mass

2. 🧱 Matter takes up space. Gen ed #10

A. False | B. Only solids do | **C. True**

3. 🍎 What type of energy can all living things use directly? Gen ed #16

A. Chemical energy | B. Light energy | C. Mechanical energy

4. 🪨 A boulder at the edge of a cliff has… From review

A. Kinetic energy | B. No energy | **C. Potential energy**

5. 🧈 Gram for gram, lipids give more energy than glucose because… Gen ed #19

A. They have more high-energy C-H and C-C bonds | B. They are lighter | C. They have more oxygen

6. 🐣 A good fuel organism makes ___ offspring each year. Gen ed #1

A. Few (low) | **B. Many (high)** | C. Zero

7. 🌾 A good fuel organism needs ___ fertilizer. Gen ed #2

A. A lot (high) | **B. Little (low)** | C. Only salt

8. 🚌 Burning biodiesel: chemical energy → ___ → mechanical. Gen ed #22

A. Heat | B. Light | C. Matter

9. 💨 When fuel burns, it uses oxygen and releases… Gen ed #23

A. Oxygen and sugar | B. Only smoke | **C. Water and carbon dioxide**

10. 🔥 In an exothermic reaction, energy is… Gen ed #24

A. Released (it gets hot) | B. Taken in (it gets cold) | C. Not changed

11. ♻️ Which fuel is a renewable biofuel? Gen ed #26

A. Coal | B. Gasoline | **C. Ethanol**

12. 🌿 Which process does NOT put carbon dioxide into the air? Gen ed #27

A. Burning | **B. Photosynthesis** | C. Breathing

13. 🔄 Algae biofuel keeps the carbon cycle balanced because carbon dioxide is… Gen ed #28

A. Only made | **B. Used and made** | C. Only used

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A · 11-C · 12-B · 13-B

Reworded from the gen ed “Algae Biofuel Final Exam (Canvas quiz, 29 Qs) + A2.10 Algae Biofuel Review”. Score: __ / 13.

🔑 *Questions tagged “Gen ed #” use the answers stored in her Canvas Algae Biofuel Final Exam (reworded). “From review” = written by our team from her A2.10 review, please check. Her paper Algae Test (32 pts) is not in the export; this modified test stands in for it.*